

The mechanism: one added term in one attention layer

1. Slots

learned + fixed

M embeddings $S \in \mathbb{R}^{M \times D}$ on a coarse grid of fixed mm coordinates $\{p_m\}$ over the volume ($M = 6^3 = 216$).

2. The gather

one cross-attention layer

Queries = slots. Keys/values = frozen vision-tower patches from all views, concatenated ($P = V \times 324$).

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}} + \frac{1}{\tau} \log w\right)$$

$w_{m,j}$: does patch j image the point p_m ? From the validated geometry kernel — *not learned*.

3. Conditioning tensor

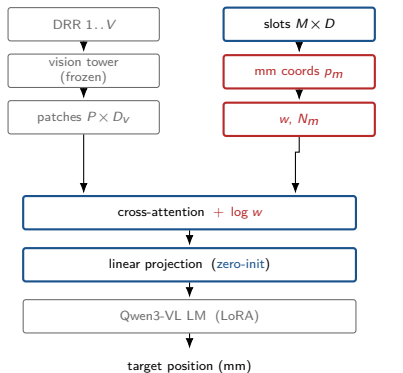
12 numbers/slot

$$N_m = \sum_v w_{m,v} d_{m,v} d_{m,v}^T, \quad d_{m,v} = \widehat{p_m - s_v}$$

Eigendecompose the 3×3 ; the 3 eigenvalues and 3 eigenvectors go through a small MLP into the slot.

4. Wiring

Slots $\rightarrow$ zero-init linear $\rightarrow$ prepended to Qwen3-VL in place of image tokens. LoRA the LM, freeze the tower.



red = geometry, fixed blue = learned grey = frozen

Why the two geometric terms do different jobs

$\log w$ restricts *where* a slot may look.

A slot at p_m can only attend to patches that physically image p_m . The forward operator is imposed, not learned.

w density (2 views)	1.9% of 648 patches
attention off-support	0.00e+00
effective width $\tau=4$	36.4 patches
$\tau=1$	12.4 patches
$\tau=0.25$	4.8 patches
uniform w (ablation)	648 patches

τ anneals $4 \rightarrow 1$: sharp from step one gives dead gradients. A slot no view can see is gated to zero — otherwise softmax over an all-masked row is *uniform* and the slot would inject an average of the whole image.

Cost. w depends only on geometry, so it is computed once per (volume, view-set) and cached — never during training.

N_m tells the slot *what its evidence cannot say*.

$N_m = \sum_v w_{m,v} d_{m,v} d_{m,v}^\top$ is an outer-product sum over the rays that image p_m , so its spectrum counts independent directions of evidence:

views	eigenvalues	rank
1 (PA)	[1, 0, 0]	1
2 (PA + LAT)	[1.02, 0.98, 0]	2

For one view the dominant eigenvector *is* the ray ($|\cos| = 1.000000$): the direction the view cannot resolve. Adding an orthogonal view removes it.

Why this matters. In-plane position lies in the *range* space of the projection and is recoverable from one view; depth along the ray lies in the *null* space and is not. N_m hands the slot that distinction explicitly, rather than asking it to infer the geometry from pixels.

Gate before any claim. FOA vs. uniform w , same data, same budget. If $\log w$ does not beat uniform, the mechanism is not doing the work.